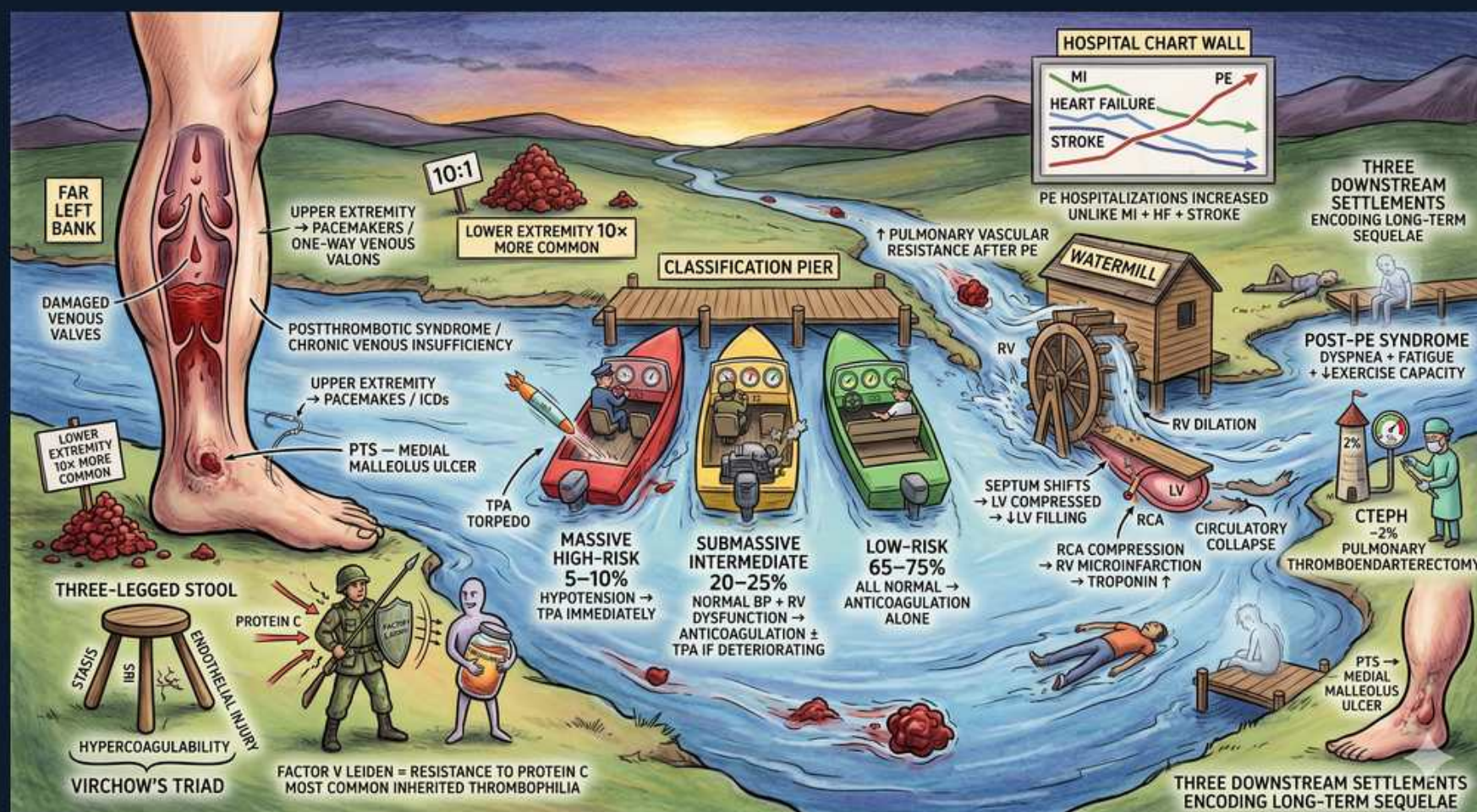


VTE — Overview, PTS, Virchow's Triad & PE Severity



1. Damaged venous valves

WHAT YOU SEE

Swollen red leg with damaged one-way valves

WHAT IT ENCODES

Deep vein thrombosis injures valves; incompetent valves cause venous reflux and chronic venous insufficiency. DVT is the proximate event for both PE and post-thrombotic syndrome. Proximal (popliteal/femoral/iliac) DVT carries far higher embolization risk than isolated distal calf DVT.

BOARD TRAP

Superficial thrombophlebitis is NOT VTE and rarely embolizes — the board wants DEEP vein involvement before anticoagulating for PE risk.

2. Lower extremity 10x more common

WHAT YOU SEE

10:1 ratio sign over the leg

WHAT IT ENCODES

Lower-extremity DVT is ~10x more common than upper-extremity DVT. Upper-extremity DVT is tied to indwelling catheters, pacemakers/ICDs, and effort thrombosis (Paget-Schroetter). Lower limb is the dominant PE source.

BOARD TRAP

New upper-extremity DVT with no catheter → think thoracic outlet (Paget-Schroetter) in a young athlete, not simply 'rare site'.

3. Upper extremity — catheters/pacemakers

WHAT YOU SEE

Pacemaker/ICD leads in upper-extremity vein

WHAT IT ENCODES

Catheter-associated and device-lead thrombosis are the leading causes of upper-extremity DVT. Management favors anticoagulation with catheter retention if still needed and functional, rather than automatic removal.

BOARD TRAP

Don't reflexively pull a still-needed central line for catheter-associated DVT — anticoagulate and keep the line if functional.

4. Post-thrombotic syndrome (PTS)

WHAT YOU SEE

Chronic venous insufficiency, brown skin changes

WHAT IT ENCODES

PTS follows 20-50% of proximal DVTs: chronic leg pain, edema, hyperpigmentation and venous stasis ulcers. Graduated compression stockings and early ambulation reduce its burden. It reflects valvular damage, not recurrent clot.

BOARD TRAP

PTS is a chronic SEQUELA, not acute recurrent DVT — new unilateral swelling years later still warrants Doppler, but PTS itself is not re-treated with full anticoagulation.

5. Medial malleolus ulcer

WHAT YOU SEE

Ulcer at the medial malleolus

WHAT IT ENCODES

Venous stasis ulcers classically sit over the medial malleolus (gaiter area) with irregular borders and exudate. Contrast with arterial ulcers (lateral/distal, punched-out, painful) and neuropathic ulcers (plantar pressure points).

BOARD TRAP

Medial malleolus + brown skin = VENOUS ulcer (compression). A lateral, punched-out, painful ulcer with absent pulses = ARTERIAL — compression would worsen it.

6. Virchow's triad

WHAT YOU SEE

Three-legged stool: stasis, endothelial injury, hypercoagulability

WHAT IT ENCODES

Virchow's triad — venous stasis, endothelial injury, and hypercoagulability — underlies all thrombosis. Each clinical risk factor maps to one leg (immobility=stasis, surgery/trauma=injury, malignancy/thrombophilia=hypercoagulability).

BOARD TRAP

All three legs need not be present — a single strong factor (e.g., active cancer) suffices to drive VTE risk.

7. Factor V Leiden / Protein C

WHAT YOU SEE

Protein C figure with Factor V Leiden label

WHAT IT ENCODES

Factor V Leiden is the most common INHERITED thrombophilia: a mutated factor V resists cleavage by activated protein C (APC resistance). It chiefly raises venous, not arterial, risk. Heterozygotes have modest risk; homozygotes much higher.

BOARD TRAP

Factor V Leiden causes APC RESISTANCE — don't confuse with protein C DEFICIENCY. Both are venous, but the mechanism (resistant substrate vs. missing enzyme) is the discriminator.

8. PE chart wall — MI/HF/stroke

WHAT YOU SEE

Wall chart trending MI, heart failure, stroke up after PE

WHAT IT ENCODES

Untreated/recurrent PE drives RV pressure overload, raising long-term cardiovascular morbidity. Acute PE classically presents with pleuritic chest pain, dyspnea, and tachycardia; massive PE causes obstructive shock.

BOARD TRAP

Sudden dyspnea + hypoxia + clear lungs + tachycardia = PE until proven otherwise — don't anchor on 'atypical MI'.

9. Massive high-risk (5-10%)

WHAT YOU SEE

Red truck, 'massive high-risk 5-10%'

WHAT IT ENCODES

Massive (high-risk) PE = sustained hypotension (SBP <90 or drop ≥40 for ≥15 min) or shock. It accounts for ~5-10% of PE and demands immediate reperfusion — systemic thrombolysis (tPA) is first-line if no contraindication.

BOARD TRAP

Hypotension defines MASSIVE PE — give thrombolysis. RV strain on echo WITHOUT hypotension is submassive (intermediate), not massive.

10. Submassive intermediate (20-25%)

WHAT YOU SEE

Yellow vehicle, 'submassive intermediate 20-25%'

WHAT IT ENCODES

Submassive (intermediate-risk) PE = normotensive but with RV dysfunction (echo/CT) and/or positive troponin/BNP. Anticoagulate; consider thrombolysis only if the patient deteriorates ('clinical worsening').

BOARD TRAP

Don't thrombolysse every submassive PE — reserve lytics for hemodynamic deterioration; routine anticoagulation otherwise.

11. Low-risk (65-75%)

WHAT YOU SEE

Green boat, 'low-risk 65-75%'

WHAT IT ENCODES

Most PEs (65-75%) are low-risk: normotensive, no RV dysfunction, negative biomarkers. Anticoagulation alone is sufficient; selected low-risk patients (PESI/sPESI low) may be treated as outpatients.

BOARD TRAP

Low-risk PE = anticoagulation ONLY — adding thrombolysis adds bleeding risk with no benefit.

12. Septum shifts → ↓LV filling

WHAT YOU SEE

Watermill, RV dilation pushing septum toward LV

WHAT IT ENCODES

Acute RV pressure overload dilates the RV and bows the interventricular septum into the LV, reducing LV preload and cardiac output — the mechanism of obstructive shock in massive PE.

BOARD TRAP

In PE shock the problem is RV failure/septal shift, not LV failure — fluids must be cautious; aggressive volume worsens RV distension.

13. RCA compression → RV ischemia

WHAT YOU SEE

RCA compression, 'RV ischemia / ↑troponin'

WHAT IT ENCODES

RV wall tension plus systemic hypotension reduces right coronary perfusion, causing RV ischemia and troponin leak — a poor-prognosis marker in PE. This is demand ischemia, not coronary thrombosis.

BOARD TRAP

Positive troponin in PE is RV strain — don't misroute the patient to cardiac cath for 'NSTEMI'.

14. CTEPH

WHAT YOU SEE

Pulmonary thromboendarterectomy figure

WHAT IT ENCODES

Chronic thromboembolic pulmonary hypertension (CTEPH) develops in ~2-4% after PE from unresolved organized clot, causing progressive dyspnea and pulmonary HTN. Pulmonary thromboendarterectomy is potentially curative; lifelong anticoagulation is required.

BOARD TRAP

Persistent dyspnea/PH months after PE → V/Q scan for CTEPH (preferred screen over CTPA), then refer for endarterectomy — not just 'deconditioning'.

15. Circulatory collapse

WHAT YOU SEE

Figure in circulatory collapse

WHAT IT ENCODES

The terminal pathway of massive PE: RV failure → low CO → obstructive shock and arrest. PEA arrest with risk factors should prompt empiric consideration of thrombolysis.

BOARD TRAP

Cardiac arrest with high PE suspicion can justify EMPIRIC thrombolysis even without confirmatory imaging.
